

$$(m - Y_{\dot{v}})\dot{v} - (mu_0 - Y_r)r - Y_v v = Y_{\delta}\delta + Y_d$$

$$(I_z - N_{\dot{r}})\dot{r} - N_v v - N_r r = N_{\delta}\delta + N_d$$

State variables: sway velocity v , yaw rate r , rudder angle δ

Hydrodynamic terms: Y_{δ} , N_{δ} , Y_d , N_d